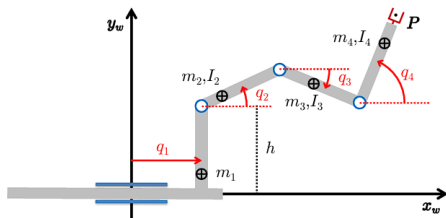


# Exercise #1

Consider the 4-dof robot in Fig. 1, made by a 3R planar arm mounted on a rail. The robot has the last three links of equal length  $\ell$ . The generalized coordinates  $q \in \mathbb{R}^4$  to be used are also shown. Determine the inertia matrix  $M(q)$  of the dynamic model of this robot (if needed, define symbolically any missing parameters).



\* we denote

$$\omega_i^T I_i \omega_i = \underbrace{I_{i2}}_{I_i} \omega_i^T \omega_i$$

we drop the  $\omega$

First: define the CoM's of the 4 links:

$$p_{c1} = \begin{cases} q_1 \\ d_{c1} \end{cases} \quad p_{c2} = \begin{cases} q_1 + c_2 d_{c2} \\ h + s_2 d_{c2} \end{cases} \quad p_{c3} = \begin{cases} q_1 + \ell c_2 + c_3 d_{c3} \\ h + \ell s_2 + s_3 d_{c3} \end{cases} \quad p_{c4} = \begin{cases} q_1 + \ell(c_2 + c_3) + c_4 d_{c4} \\ h + \ell(s_2 + s_3) + s_4 d_{c4} \end{cases}$$

the angular speed of the links is:

$$\omega_1 = 0 \quad \omega_2 = \dot{q}_2 \quad \omega_3 = \dot{q}_3 \quad \omega_4 = \dot{q}_4$$

$$v_1 = (\dot{q}_1, 0)^T \Rightarrow T_1 = \frac{1}{2} m_1 \dot{q}_1^2$$

$$v_2 = (\dot{q}_1 - d_{c2} s_2 \dot{q}_2, d_{c2} c_2 \dot{q}_2)^T \Rightarrow \frac{1}{2} [m_2 (\dot{q}_1^2 + d_{c2}^2 \dot{q}_2^2 - 2 d_{c2} s_2 \dot{q}_1 \dot{q}_2) + I_2 \dot{q}_2^2] =$$

$$v_3 = \begin{cases} \dot{q}_1 - \ell s_2 \dot{q}_2 - d_{c3} s_3 \dot{q}_3 \\ \ell c_2 \dot{q}_2 + d_{c3} c_3 \dot{q}_3 \end{cases} \quad T_2 = \frac{1}{2} [m_2 \dot{q}_1^2 - 2 m_2 d_{c2} s_2 \dot{q}_1 \dot{q}_2 + (I_2 + m_2 d_{c2}^2) \dot{q}_2^2]$$

$$T_3 = \frac{1}{2} m_3 (\dot{q}_1^2 + \ell^2 s_2^2 \dot{q}_2^2 + d_{c3}^2 s_3^2 \dot{q}_3^2 - 2 (\ell s_2 \dot{q}_1 \dot{q}_2 + d_{c3} s_3 \dot{q}_1 \dot{q}_3 - \ell s_2 s_3 d_{c3} \dot{q}_2 \dot{q}_3) + \ell^2 c_2^2 \dot{q}_2^2 + d_{c3}^2 c_3^2 \dot{q}_3^2 + 2 \ell d_{c3} c_2 c_3 \dot{q}_2 \dot{q}_3) + \frac{1}{2} I_3 \dot{q}_3^2$$

$$= \frac{1}{2} [m_3 \dot{q}_1^2 + (m_3 \ell^2) \dot{q}_2^2 + (m_3 d_{c3}^2 + I_3) \dot{q}_3^2 - 2 \ell s_2 m_3 \dot{q}_1 \dot{q}_2 - 2 d_{c3} s_3 m_3 \dot{q}_1 \dot{q}_3 + \cos(q_2 - q_3) \ell d_{c3} m_3 \dot{q}_2 \dot{q}_3]$$

$$v_4 = \begin{cases} \dot{q}_1 - \ell (s_2 \dot{q}_2 + s_3 \dot{q}_3) - s_4 d_{c4} \dot{q}_4 \\ \ell (c_2 \dot{q}_2 + c_3 \dot{q}_3) + c_4 d_{c4} \dot{q}_4 \end{cases}$$

$$v_4^T v_4 = \dot{q}_1^2 + \ell^2 s_2^2 \dot{q}_2^2 + \ell^2 s_3^2 \dot{q}_3^2 + 2 \ell^2 s_2 s_3 \dot{q}_2 \dot{q}_3 + s_4^2 d_{c4}^2 \dot{q}_4^2 - 2 \dot{q}_1 \ell (s_2 \dot{q}_2 + s_3 \dot{q}_3) - 2 \dot{q}_1 s_4 d_{c4} \dot{q}_4 + 2 \ell (s_2 \dot{q}_2 + s_3 \dot{q}_3) s_4 d_{c4} \dot{q}_4 + \ell^2 c_2^2 \dot{q}_2^2 + \ell^2 c_3^2 \dot{q}_3^2 + 2 \ell^2 c_2 c_3 \dot{q}_2 \dot{q}_3 + c_4^2 d_{c4}^2 \dot{q}_4^2 + 2 \ell (c_2 \dot{q}_2 + c_3 \dot{q}_3) c_4 d_{c4} \dot{q}_4$$

$$\Rightarrow T_4 = \frac{1}{2} I_4 \dot{q}_4^2 + \frac{1}{2} m_4 \left[ \dot{q}_1^2 + \ell^2 \dot{q}_2^2 + 2 \ell^2 \cos(q_2 - q_3) \dot{q}_2 \dot{q}_3 + d_{c4}^2 \dot{q}_4^2 - 2 \ell s_2 \dot{q}_1 \dot{q}_2 - 2 \ell s_3 \dot{q}_1 \dot{q}_3 - 2 s_4 d_{c4} \dot{q}_1 \dot{q}_4 + 2 \ell d_{c4} (\cos(q_2 - q_4) \dot{q}_2 \dot{q}_4 + \cos(q_3 - q_4) \dot{q}_3 \dot{q}_4) + \ell^2 \dot{q}_2^2 \right]$$

$$M_{11} = m_1 + m_2 + m_3 + m_4$$

$$M_{12} = -(m_2 d_{c2} + \ell(m_3 + m_4)) S_2$$

$$M_{13} = -(d_{c3} m_3 + \ell m_4) S_3$$

$$M_{14} = S_4 d_{c4} m_4$$

$$M_{22} = I_2 + m_2 d_{c2}^2 + \ell^2 (m_3 + m_4)$$

$$M_{23} = \cos(q_2 - q_3) \ell d_{c3} m_3 + \ell^2 \cos(q_2 - q_3) m_4$$

$$M_{24} = \ell d_{c4} \cos(q_2 - q_4)$$

$$M_{33} = m_3 d_{c3}^2 + I_3 + \ell^2 m_4$$

$$M_{34} = \cos(q_3 - q_4) \ell d_{c4}$$

$$M_{44} = m_4 d_{c4}^2 + I_4$$

## Exercise #2

For the same robot in Fig. 1, assume  $\ell = h = 1$  [m] and consider the following tasks, to be executed using a kinematic control scheme with joint velocity commands  $\dot{q} \in \mathbb{R}^4$ .

**Task 1.** Trace with the end-effector counterclockwise a circle of radius  $R = 3$  [m], centered at  $C = (7, 0)$ , starting from point  $P_0 = (10, 0)$  and with constant speed  $v = 1$  [m/s].

**Task 2.** Keep the second link always horizontal ( $q_2(t) = 0$ ).

Define the augmented Jacobian  $J_A(q)$  for both tasks 1 and 2. Choose a suitable initial robot configuration so as to stay at time  $t = 0$  in  $P_0$  and compute there the minimum joint velocity norm solution that realizes both tasks simultaneously. Determine the first point  $P_s$  on the circular path where an algorithmic singularity of  $J_A(q)$  necessarily occurs. In that situation, compute the minimum joint velocity norm solution that realizes the first task only. Will the execution of the second task be relaxed or not?

The first task is relative to ee position so

$$\begin{aligned} \xi_1: \begin{cases} q_1 + c_2 + c_3 + c_4 \\ 1 + s_2 + s_3 + s_4 \end{cases} &\Rightarrow J_1 = \begin{pmatrix} 1 & -s_2 & -s_3 & -s_4 \\ 0 & c_2 & c_3 & c_4 \end{pmatrix} \\ \xi_2 = q_2 &\Rightarrow J_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \end{pmatrix} \end{aligned} \quad \left. \vphantom{\begin{matrix} J_1 \\ J_2 \end{matrix}} \right\} J_A$$

An initial suitable conf. to stay in  $P_0 = (10, 0)$  is  
 $q = (10, 0, -\pi/2, \pi)^T$

$$\xi_d = \begin{cases} 7 + 3 \cos(\frac{1}{3}t) \\ 3 \sin(\frac{1}{3}t) \\ 0 \end{cases} \Rightarrow \dot{\xi}_d = \begin{cases} -\sin(t) \\ \cos(t) \\ 0 \end{cases} \Rightarrow \dot{\xi}_d(0) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow J_A(q_0) = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \Rightarrow \dot{q}_0 = J_A^\#(q_0) \dot{\xi}_d(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

$$J_A = \begin{pmatrix} 1 & -s_2 & -s_3 & -s_4 \\ 0 & c_2 & c_3 & c_4 \\ 0 & 1 & 0 & 0 \end{pmatrix} \text{ a singularity is reached when } \xi^* = (7, 3, 0)$$

$$\Rightarrow q^* = (6, 0, \pi/2, \pi/2) \Rightarrow J_A(q^*) = \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \Rightarrow \text{rank } J_A = 2 < 3!$$

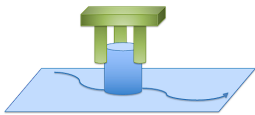
the solution to realize  $\xi_1$  is

$$\dot{q} = J_1^\# \xi_1 \left( \frac{\pi}{2} \right) = \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix}^\# \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{3} & -\frac{1}{3} \end{pmatrix}^T$$

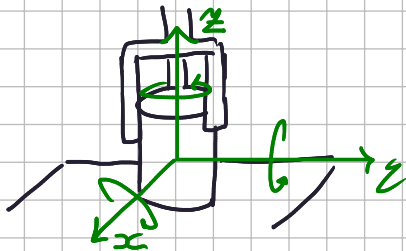
$\Rightarrow$  execution of second task is not relaxed.

### Exercise #3

A 6-dof robot should hold firmly with its three-fingered gripper a cylindric payload, and move it along a desired path on a frictionless plane with one of its bases in full contact with the plane, as shown in Fig. 2. Define an associated task frame where the natural (geometric) constraints and the artificial (control) constraints of this hybrid task can be defined and realized in a decoupled way. Where reasonable, provide also values for the control references.



We define the task frame  $RF_t$  with  $(x, y)$  parallel to the plane.



in this way the natural constraints are

$$\begin{cases} v_z = 0 & \omega_y = 0 & \omega_x = 0 \\ f_x = 0 & f_y = 0 & \mu_z = 0 \end{cases} \text{ so:}$$

$$\text{So we set } t = \begin{pmatrix} v \\ \omega \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ \omega_z \end{pmatrix} = S_t \overset{\text{desired par. twist}}{V}$$

So we can control only the reactions  $S_z, \mu_x, \mu_y$ :

$$\omega = \begin{pmatrix} f \\ \mu \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} S_z \\ \mu_x \\ \mu_y \end{pmatrix} = S_w \lambda \text{ the control reference given}$$

$$\text{as artificial constr. are } V = \begin{pmatrix} v_{xd}(t) \\ v_{yd}(t) \\ 0 \end{pmatrix} \quad \lambda = \begin{pmatrix} S_{zd} \\ 0 \\ 0 \end{pmatrix} \leftarrow \text{to keep contact.}$$

Exercise #4

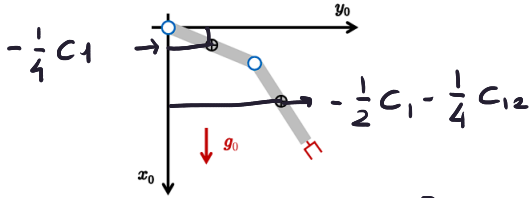
During the accurate execution of a smooth joint trajectory  $q_d(t)$  lasting  $T = 3$  [s] with the 2R planar robot shown in Fig. 3 moving under gravity, the maximum torques of the two joints exceed at some instants their bounds, as given by  $|\tau_i| \leq \tau_{max,i}$ ,  $i = 1, 2$ . We have in particular

$\tau_1(t_1) = \max_{t \in [0,T]} \tau_1(t) = 140 > 100 = \tau_{max,1}$ ,  $\tau_2(t_2) = \max_{t \in [0,T]} \tau_2(t) = 25 > 20 = \tau_{max,2}$  [Nm].

The robot links have equal length  $\ell = 0.5$  [m] and equal, uniformly distributed mass  $m = 5$  [kg]. The robot configurations at the time instants  $t = t_1$  and  $t = t_2$  are

$q(t_1) = (\pi/4 \ 0)^T$ ,  $q(t_2) = (-\pi/4 \ 3\pi/4)^T$  [rad].

In order to recover motion feasibility, a uniform trajectory scaling is used. What will be the minimum feasible motion time  $T' = kT$  thus obtained? What are the values of the new joint torques  $\tau_1(kt_1)$  and  $\tau_2(kt_2)$ ?



I derive the gravity term:  $U = -g_0 m (\frac{1}{4} C_1 + \frac{1}{2} C_1 + \frac{1}{4} C_{12}) = -4g (\frac{3}{4} C_1 + \frac{1}{4} C_{12})$

$\Rightarrow g = \frac{\partial U}{\partial q} : \begin{cases} 4g (\frac{3}{4} s_1 + \frac{1}{4} s_{12}) \\ \frac{4g}{4} s_{12} \end{cases} \Rightarrow \begin{cases} g_1(t_1) = 34.6482 \\ g_2(t_2) = 12.25 \end{cases}$

$\Rightarrow K' = \max \left\{ 1, \frac{140 - 34.6482}{100 - 34.6482}, \frac{25 - 12.25}{20 - 12.25} \right\} = \max \{ 1.612, 1.6451 \} \Rightarrow K = \sqrt{K'} = 1.2826$

$\Rightarrow q'(r) = q(Kt) \Rightarrow T' = 3.847$

$\begin{cases} z_1(Kt_1) = \frac{1}{1.6451} (140 - 34.6482) + 34.6482 \approx 99 \\ z_2(Kt_2) = \frac{1}{1.6451} (25 - 12.25) + 12.25 \approx 17 \end{cases}$

Exercise #5

Consider again the 2R robot in Fig. 3, with the same definition of joint variables and using the same kinematic and dynamic parameters. The robot is initially at rest at  $t = 0$  in  $q(0) = q_0$ . Provide the explicit expressions of all terms in the following three feedback control laws, each achieving its own objective.

- a. Global exponential stabilization of the state  $(q, \dot{q}) = (q_d, 0)$ , with decoupled transient evolutions of the position errors  $e_i(t) = q_{d,i} - q_i(t)$ ,  $i = 1, 2$ , of the form  $e_1(t) = e_1(0) (1 + 5t) \exp(-5t)$  and  $e_2(t) = e_2(0) (2 \exp(-5t) - \exp(-10t))$ .
- b. Global asymptotic stabilization of the state  $(q, \dot{q}) = (q_d, 0)$ , without knowledge of the robot inertia matrix.
- c. Exponential stabilization of the end-effector position  $p = p_d \in \mathbb{R}^2$  with zero velocity  $\dot{p} = 0$ , up to kinematic singularities.

2) We need feedback linearization. The model is

$M\ddot{q} + C + g = \tau$  i choose  $\tau = C + g + M\alpha \Rightarrow$  the closed loop system becomes  $M\ddot{q} + C + g = C + g + M\alpha \Rightarrow \ddot{q} = \alpha$  i choose  $\alpha$  as a PD

$\ddot{q} = K_P(q - q_d) - K_D\dot{q} \Rightarrow \begin{cases} \ddot{q}_1 = K_{P1}(q_{d1} - q_1) - K_{D1}\dot{q}_1 \\ \ddot{q}_2 = K_{P2}(q_{d2} - q_2) - K_{D2}\dot{q}_2 \end{cases}$

$\Rightarrow \ddot{e}_i + K_{D_i}\dot{e}_i + K_{P_i}e_i = 0$ , to have  $e_1(t)$  i need to choose  $K_{P1}$  and  $K_{D1}$  in order to get  $(\frac{K_{P1}}{2})^{1/2} = 5 \Rightarrow K_{P1} = 50 \Rightarrow K_{D1} = 10$

to have  $e_2$  i need:

$\lambda_1 = 5, \lambda_2 = 10$

$\begin{cases} 5 = \frac{K_{D2}}{2} - \sqrt{\frac{K_{D2}^2}{4} - \frac{K_{P2}}{2}} \\ 10 = \frac{K_{D2}}{2} + \sqrt{\frac{K_{D2}^2}{4} - \frac{K_{P2}}{2}} \end{cases} \Rightarrow \begin{cases} K_{D2} = 15 \\ K_{P2} = 100 \end{cases} \Rightarrow \tau = C + g + M \left( \begin{pmatrix} 50 & 0 \\ 0 & 100 \end{pmatrix} e - \begin{pmatrix} 10 & 0 \\ 0 & 15 \end{pmatrix} \dot{q} \right)$

b) We just need a PD + gravity cancellation :  $\tau = K_p e - K_d \dot{q} + g(q)$

c) We need to perform FBL in the task space:

In the task space:

$$\underbrace{(\mathcal{J} \bar{M}^{-1} \mathcal{J}^T)^{-1}}_{M_\tau} \ddot{\zeta} - \underbrace{(\mathcal{J} \bar{M}^{-1} \mathcal{J}^T)^{-1} (\dot{\mathcal{J}} \dot{q} - \mathcal{J} \bar{M}^{-1} h)}_{n_\tau} = F$$

$\begin{matrix} \uparrow & \nearrow \\ C+g & \mathcal{J}^T \tau \end{matrix}$

we let  $\tau = \mathcal{J}^T (M_\tau a + n_\tau) \Rightarrow \ddot{\zeta} = a$ , we choose

$$a = K_p (p_d - \zeta) - K_d \dot{\zeta}$$